CASE 02 · BUSINESS ANALYST PORTFOLIO PROJECT

The $40K Verification Gap

Re-engineering a broken debt-collection process — built on the real CFPB Consumer Complaint Database.

Role simulated: Business Analyst | Tools: Python, Lucidchart-style process mapping, Excel (ROI model), AI-assisted text triage

Source: CFPB Consumer Complaint Database, Debt collection product, Jan–Mar 2015 extract (consumerfinance.gov)

45%	63%	$40K/yr	13.5 mo
of all debt-collection complaints are one issue	of that issue is “this debt isn't mine”	Modeled net savings from a verification gate	Modeled payback period

The business problem

A debt-collection process is generating a disproportionate share of regulatory complaints. Rather than start with a stakeholder-supplied hunch, this project starts from real complaint records: 7,494 debt-collection complaints filed with the CFPB in a single quarter, filtered and mined for the root cause before any process gets redesigned.

Root-cause analysis

One issue category dominates: "Cont'd attempts collect debt not owed" accounts for 3,351 of 7,494 debt-collection complaints (45%) — everything else is a long tail by comparison.

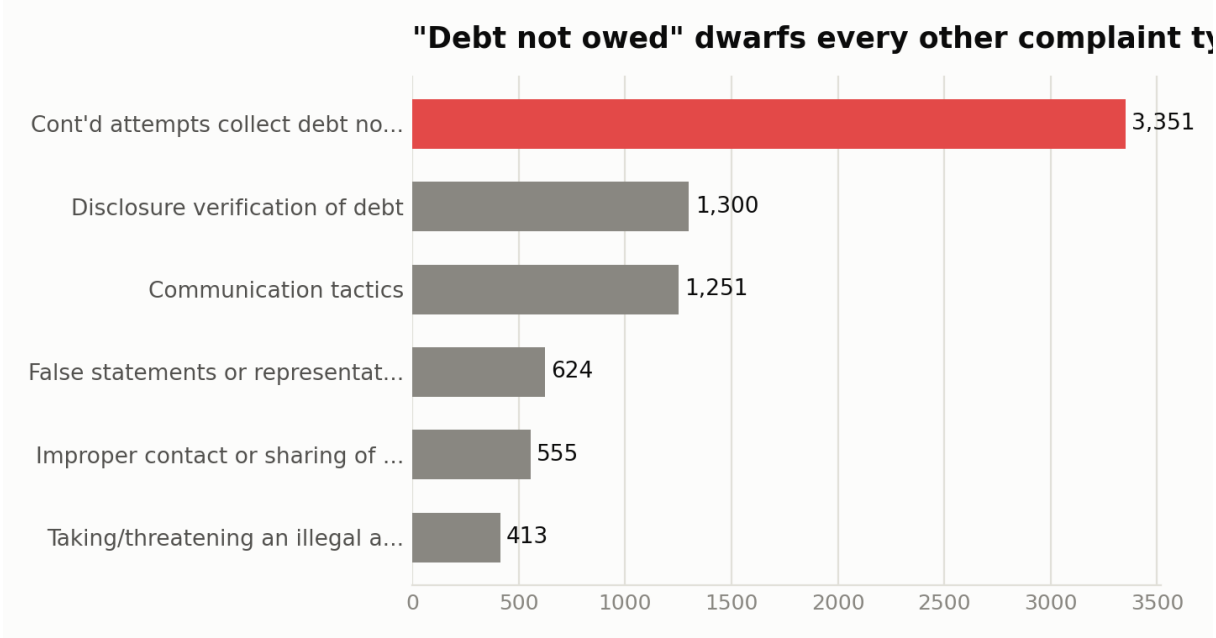


Fig. 1 — Complaint volume by issue, Debt collection product.

Drilling into that one issue's sub-categories is where the actual root cause shows up: 63% of "Cont'd attempts collect debt not owed" complaints are consumers saying **the debt isn't theirs** — not a payment dispute, not a hardship request, an identity/account-matching failure that happens before the consumer is ever contacted.

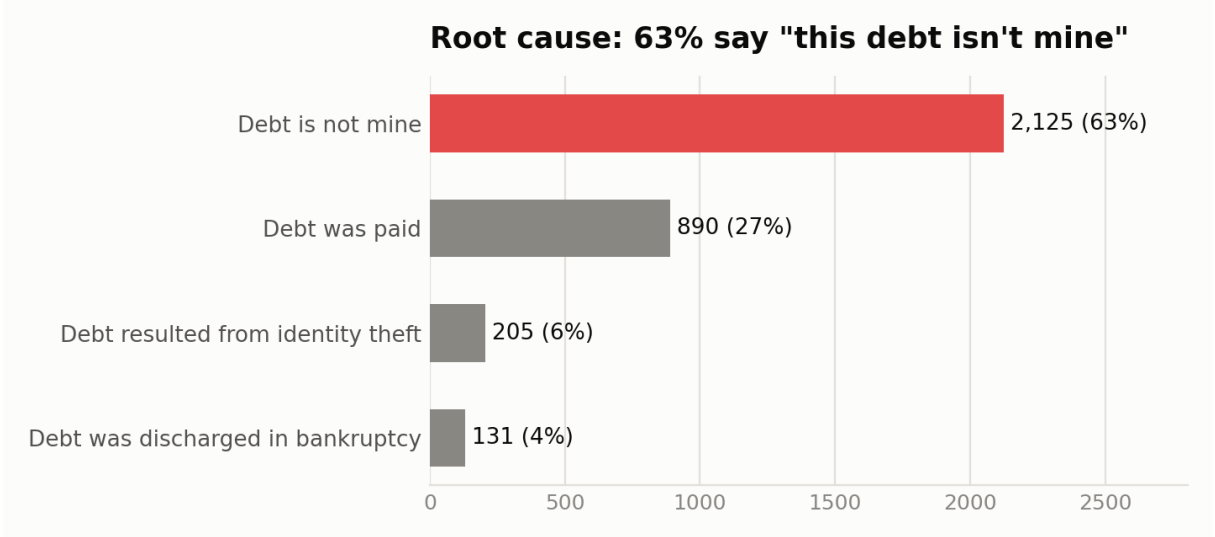


Fig. 2 — Sub-issue breakdown of the #1 complaint category.

AI checkpoint

Complaint narratives aren't included in this public extract, so the root-cause read here comes from the CFPB's own issue/sub-issue coding, not free-text AI clustering — a useful reminder that AI text-mining is only as good as the text it's given. Where a live engagement includes narrative text, the same workflow applies: have an AI assistant propose theme clusters, then hand-check a sample against the sub-issue field

already present here as ground truth before trusting the clusters.

Process redesign

The current process has no identity- or balance-verification step before a collector makes first contact — the dispute only surfaces after the consumer has already been contacted about a debt that, in a majority of these cases, isn't theirs. The fix is a gate, not a new team: verify before contacting, not after.

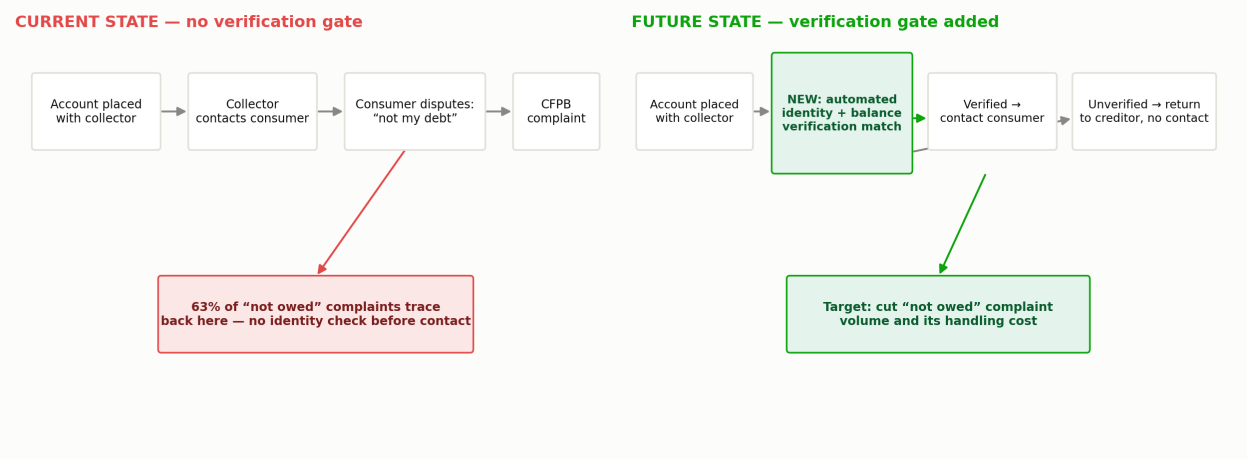


Fig. 3 — Current-state vs. future-state process map.

RACI — future-state process

Step	Responsible	Accountable	Consulted	Informed
Receive account from creditor	Intake Ops	Collections Lead	Creditor	—
Run automated identity + balance match	Verification System	Compliance/Risk	Data/IT	Collections Agent
Route verified accounts to agent	Verification System	Collections Lead	—	Collections Agent
Route unverified accounts back to creditor	Verification System	Compliance/Risk	Creditor	Collections Lead
Contact verified consumer	Collections Agent	Collections Lead	Compliance/ Risk	Consumer
Handle any remaining dispute	Collections Agent	Compliance/Risk	Legal	Consumer

Cost-benefit case

Scoped to the single highest-complaint-volume company in the sample (Enhanced Recovery Company, LLC) as a realistic company-level proxy — the full, formula-driven model ships alongside this report as **case2_roi_model.xlsx**; every number below recalculates if you change an assumption.

Input	Value	Type
"Not mine" complaints observed (74-day sample)	176	From data
Estimated annualized volume	~868 / yr	Formula
Cost to process one complaint	$150	Assumption (edit in Excel)
Expected reduction from verification gate	40%	Assumption (edit in Excel)
One-time build cost	$45,000	Assumption (edit in Excel)
Ongoing annual run cost	$12,000	Assumption (edit in Excel)
Net annual savings	$40,086	Formula
Payback period	13.5 months	Formula
3-year cumulative net benefit	$75,259	Formula

Recommendation

Add an automated identity- and balance-verification gate before first contact. It pays back in roughly 13.5 months under conservative assumptions, and — unlike most process fixes — its impact is directly traceable to a specific, quantified complaint category rather than a general efficiency argument.

RESUME BULLET, READY TO USE

“Analyzed 7,494 real CFPB debt-collection complaints to isolate a single root cause behind 44% of filings; mapped current- and future-state processes and built a cost-benefit model projecting $40K/year in net savings with a 13.5-month payback.”

What's in this package

File	What it is
run_analysis.py	Loads the CFPB extract, runs the root-cause analysis, generates the charts and process map.
case2_roi_model.xlsx	Live cost-benefit model — formulas, not hardcoded numbers. Edit the yellow cells for your own case.
This report (PDF)	Root-cause analysis, process maps, RACI, and recommendation.